13.1.4 Acidic nature of nitro alkanes

The α-H atom of 1° & 2° nitroalkanes show acidic character because of the electron with drawing effect of NO_2 group. These are more acidic than aldehydes, ketones, ester and cyanides. Nitroalkanes dissolve in NaOH solution to form a salt. Aci – nitro derivatives are more acidic than nitro form. When the number of alkyl group attached to α carbon increases, acidity decreases. due to +I effect of alkyl groups.

$$CH_3 - NO_2 > CH_3CH_2 - NO_2 > \begin{matrix} CH_3 \\ \diagdown \\ CH - NO_2 \\ \diagup \\ CH_3 \end{matrix}$$

Evaluate yourself

Write all possible isomers for the following compounds.

i) C_2H_5-NO_2 ii) C_3H_7-NO_2

13.1.5 Preparation of nitroalkanes

1) From alkyl halides: (Laboratory method)

a) Alkyl bromides (or) iodides on heating with ethanolic solution of potassium nitrite gives nitroethane.

$$\underset{\text{Ethyl bromide}}{CH_3CH_2\text{-Br}} + KNO_2 \xrightarrow[SN^2]{\text{ethanol}/\Delta} \underset{\text{Nitroethane}}{CH_3CH_2\text{-}NO_2} + KBr$$

The reaction follows SN2 mechanism.

This method is not suitable for preparing nitrobenzene because the bromine directly attached to the benzene ring cannot be cleaved easily.

2) Vapour phase nitration of alkanes: (Industrial method)

Gaseous mixture of methane and nitric acid passed through a red hot metal tube to give nitromethane.

$$\underset{\text{Methane}}{CH_{4(g)}} + HNO_{3\,(g)} \xrightarrow[\text{Red hot Si tube}]{675\ K} \underset{\text{nitro methane}}{CH_3\text{-}NO_2} + H_2O$$

Except methane, other alkanes (upto n – hexane) give a mixture of nitroalkanes due to C-C cleavage. The individual nitro alkanes can be separated by fractional distillation.

$$\underset{\text{Ethane}}{CH_3\text{-}CH_3} + HNO_3 \xrightarrow{675\ K} \underset{\text{Nitroethane(73\%)}}{CH_3CH_2\text{-}NO_2} + \underset{\text{Nitromethane (27\%)}}{CH_3NO_2}$$

3) From α- halocarboxylic acid

α-choloroacetic acid when boiled with aqueous solution of sodium nitrite gives nitromethane.

$$\underset{\alpha\text{ - chloro acetic acid}}{Cl\text{ - }CH_2\text{-COOH}} + NaNO_2 \xrightarrow[SN^2]{H_2O/Heat} \underset{\text{Nitromethane}}{CH_3\text{-}NO_2} + CO_2 + NaCl$$

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